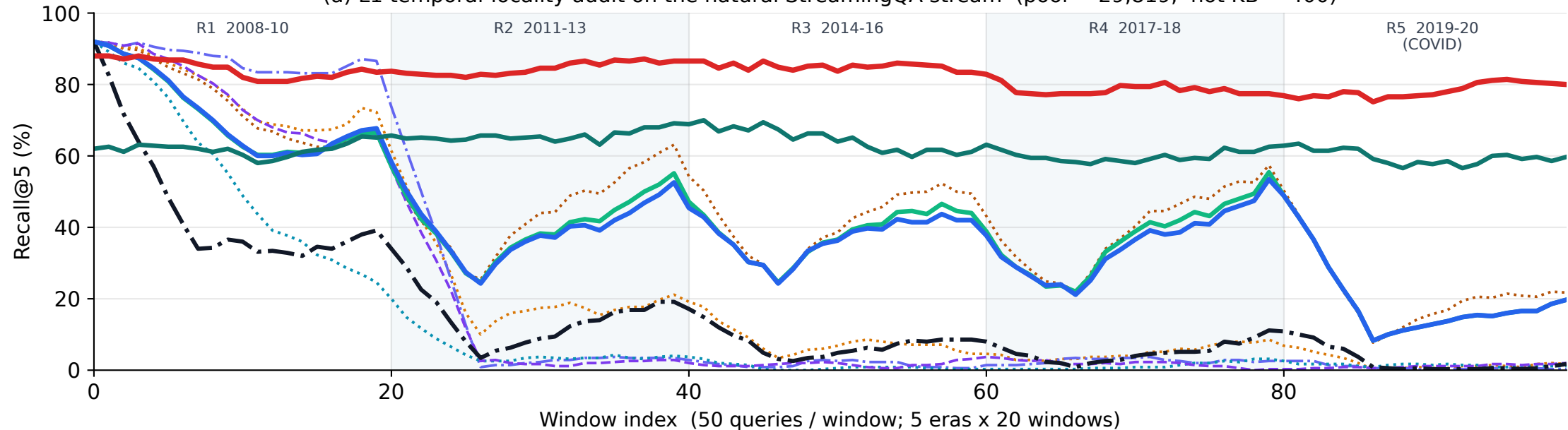
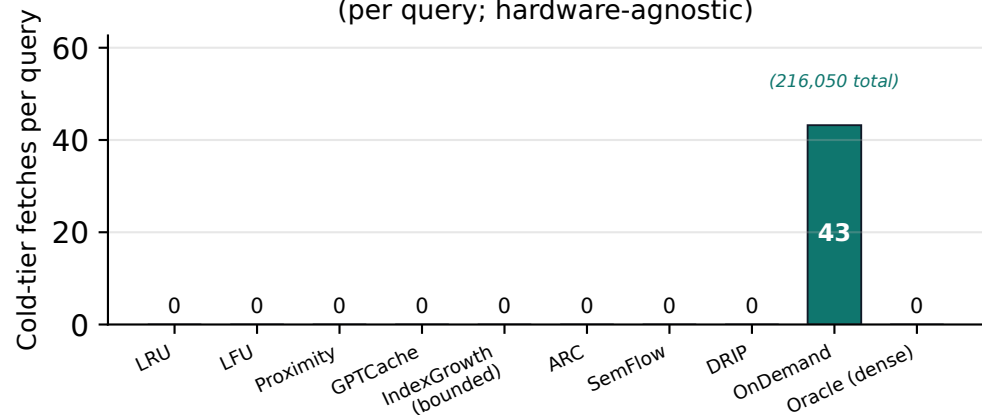


(a) L1 temporal-locality audit on the natural StreamingQA stream (pool = 29,819, hot KB = 400)



(b) Serve-time cold-tier retrieval cost (per query; hardware-agnostic)



(c) Algorithmic maintenance overhead (per window; simulated CPU, not disk I/O)

